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Course/Code : DSA0604 - DATA HANDLING AND VISUALIZATION

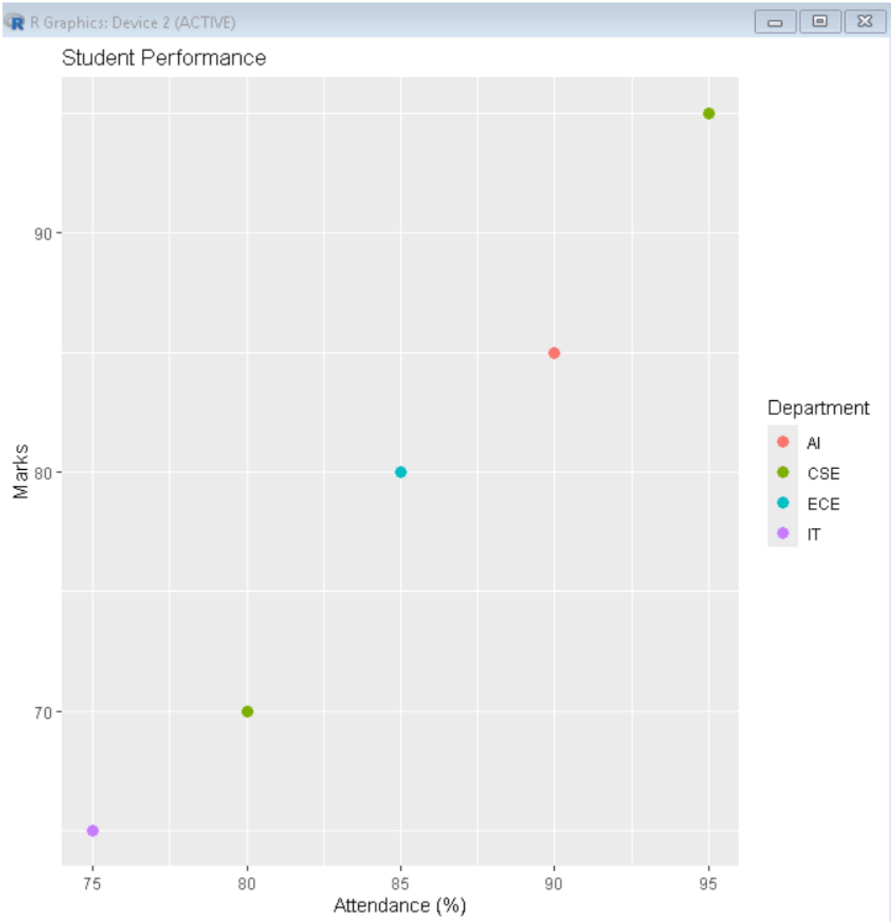
Date : 10/07/26

QUESTION 1: STUDENT ACADEMIC PERFORMANCE

CODE:

```
ggplot(student, aes(x = Attendance, y = Marks, color = Department)) +
  geom_point(size = 3) +
  labs(
    title = "Student Performance",
    x = "Attendance (%)",
    y = "Marks"
  )
```

OUTPUT:



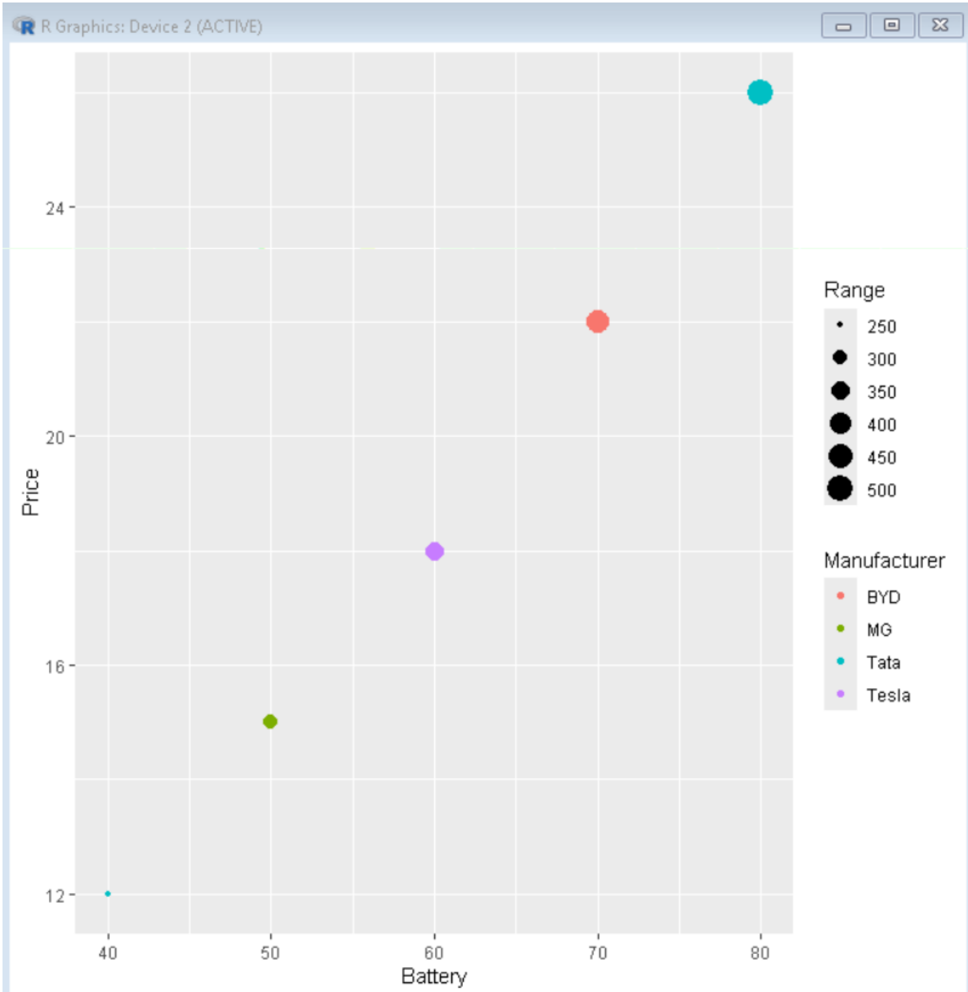
QUESTION 2: ELECTRIC VEHICLE MARKET ANALYSIS

CODE:

```
ev <- data.frame(
  Battery = c(40,50,60,70,80),
  Price = c(12,15,18,22,26),
  Range = c(250,300,350,420,500),
  Manufacturer = c("Tata","MG","Tesla","BYD","Tata")
)

ggplot(ev, aes(x=Battery, y=Price, size=Range, color=Manufacturer)) +
  geom_point()
```

OUTPUT:



QUESTION 3: HOSPITAL PATIENT ADMISSION ANALYSIS

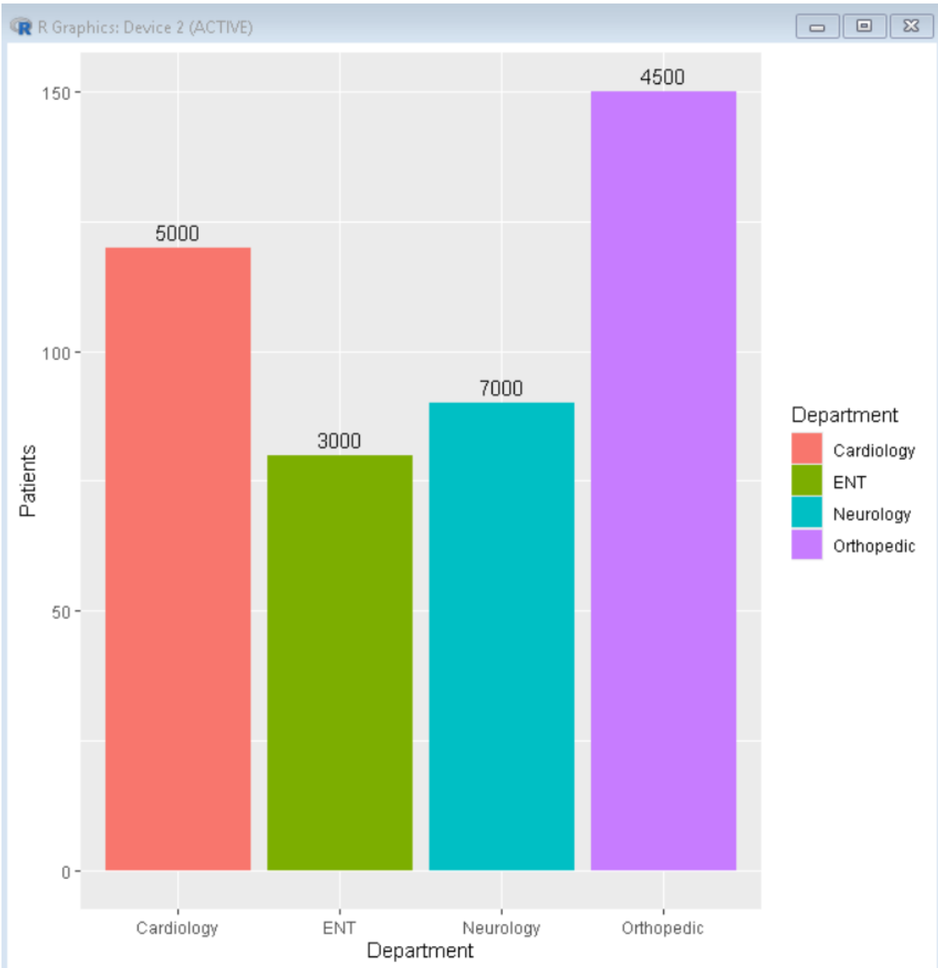
CODE:

```
hospital <- data.frame(
  Department = c("Cardiology","Neurology","Orthopedic","ENT"),
  Patients = c(120,90,150,80),
  Cost = c(5000,7000,4500,3000)
)

ggplot(hospital, aes(x=Department, y=Patients, fill=Department)) +
```

```
  geom_bar(stat="identity") +
  geom_text(aes(label=Cost), vjust=-0.5)
```

OUTPUT:



QUESTION 4: WEATHER MONITORING SYSTEM

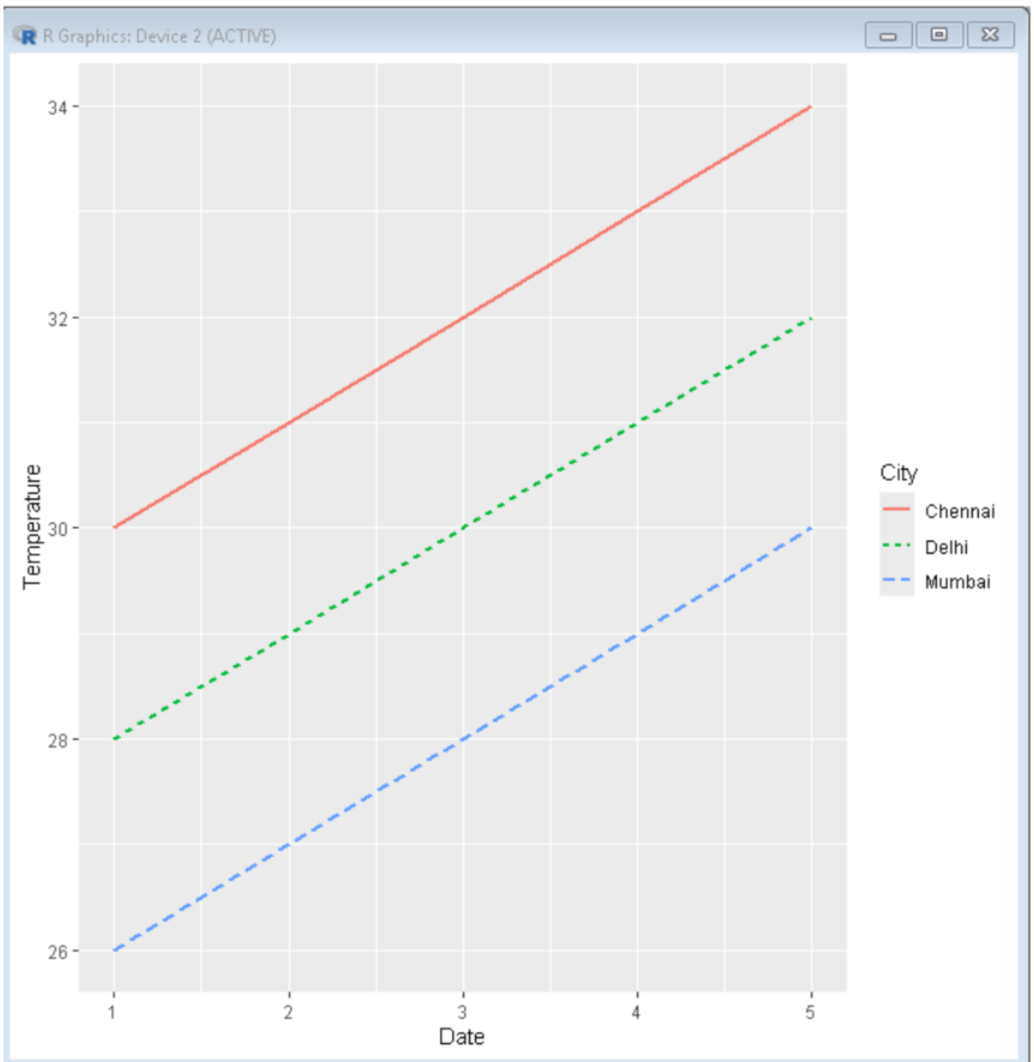
CODE:

```
weather <- data.frame(
  Date = rep(1:5,3),
```

```
  Temperature = c(30,31,32,33,34,
                  28,29,30,31,32,
                  26,27,28,29,30),
  City = rep(c("Chennai","Delhi","Mumbai"), each=5)
)

ggplot(weather,
       aes(x=Date, y=Temperature,
           color=City,
           linetype=City)) +
  geom_line(linewidth=1)
```

OUTPUT:



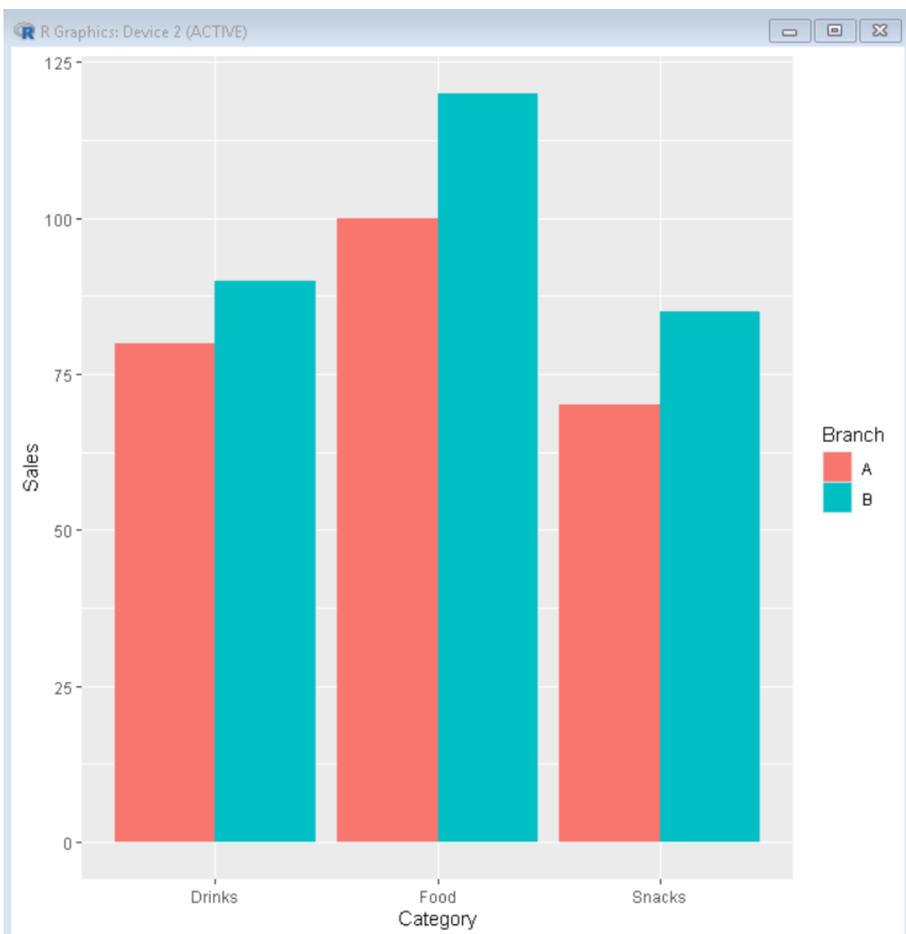
QUESTION 5: SUPERMARKET SALES ANALYSIS

CODE:

```
sales <- data.frame(
  Category = c("Food","Food","Drinks","Drinks","Snacks","Snacks"),
  Sales = c(100,120,80,90,70,85),
  Branch = c("A","B","A","B","A","B")
)

ggplot(sales,
       aes(x=Category,
           y=Sales,
           fill=Branch)) +
  geom_bar(stat="identity", position="dodge")
```

OUTPUT:

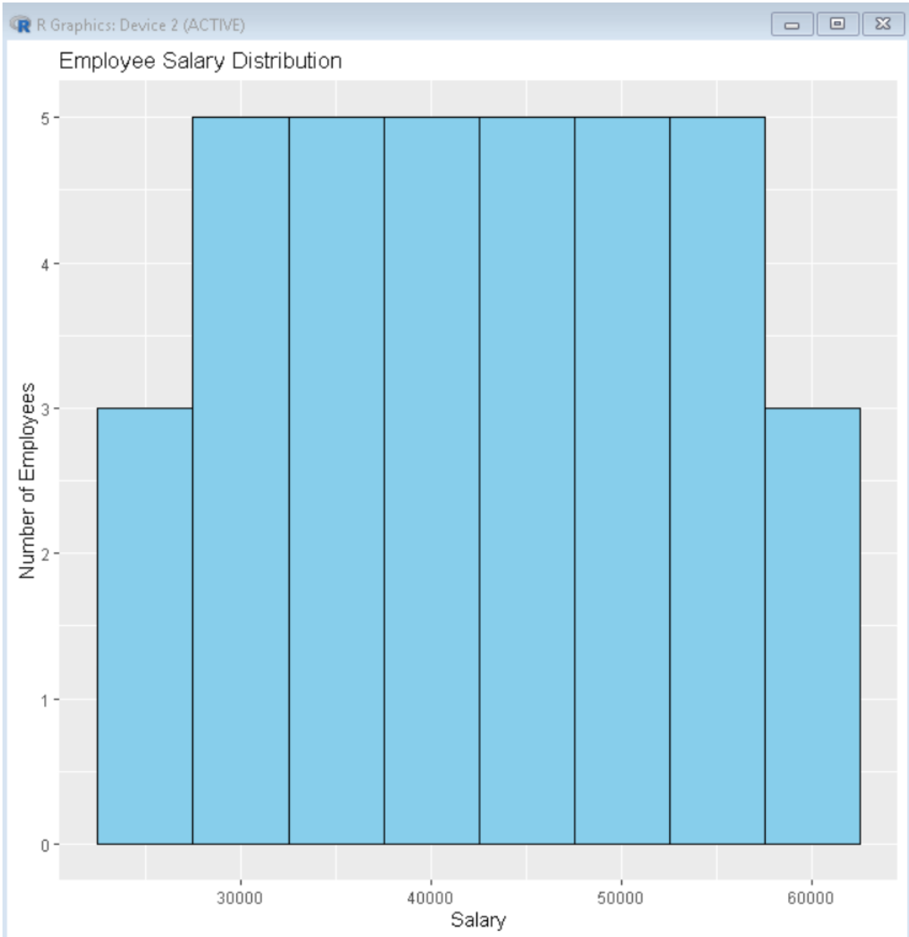


QUESTION 6: EMPLOYEE SALARY DISTRIBUTION

CODE:

```
salary <- data.frame(
  Salary = c(
    25000,26000,27000,28000,29000,
    30000,31000,32000,33000,34000,
    35000,36000,37000,38000,39000,
    40000,41000,42000,43000,44000,
    45000,46000,47000,48000,49000,
    50000,51000,52000,53000,54000,
    55000,56000,57000,58000,59000,60000
  )
)
ggplot(salary, aes(x = Salary)) +
  geom_histogram(binwidth = 5000, fill = "skyblue", color = "black") +
  labs(
    title = "Employee Salary Distribution",
    x = "Salary",
    y = "Number of Employees"
  )
```

OUTPUT:



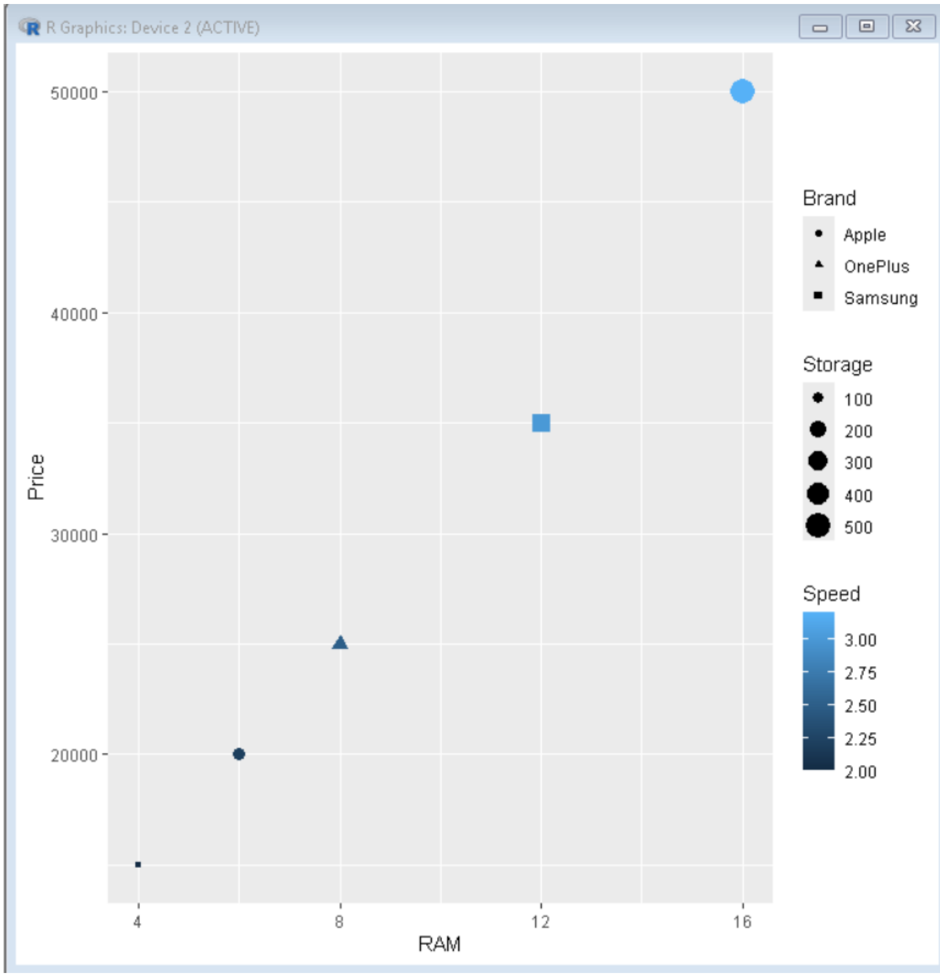
QUESTION 7: SMARTPHONE MARKET COMPARISON

CODE:

```
phone <- data.frame(
  RAM = c(4,6,8,12,16),
  Price = c(15000,20000,25000,35000,50000),
  Brand = c("Samsung","Apple","OnePlus","Samsung","Apple"),
  Storage = c(64,128,128,256,512),
  Speed = c(2.0,2.2,2.5,3.0,3.2)
)

ggplot(phone,
       aes(x=RAM,
           y=Price,
           shape=Brand,
           size=Storage,
           color=Speed)) +
  geom_point()
```

OUTPUT:



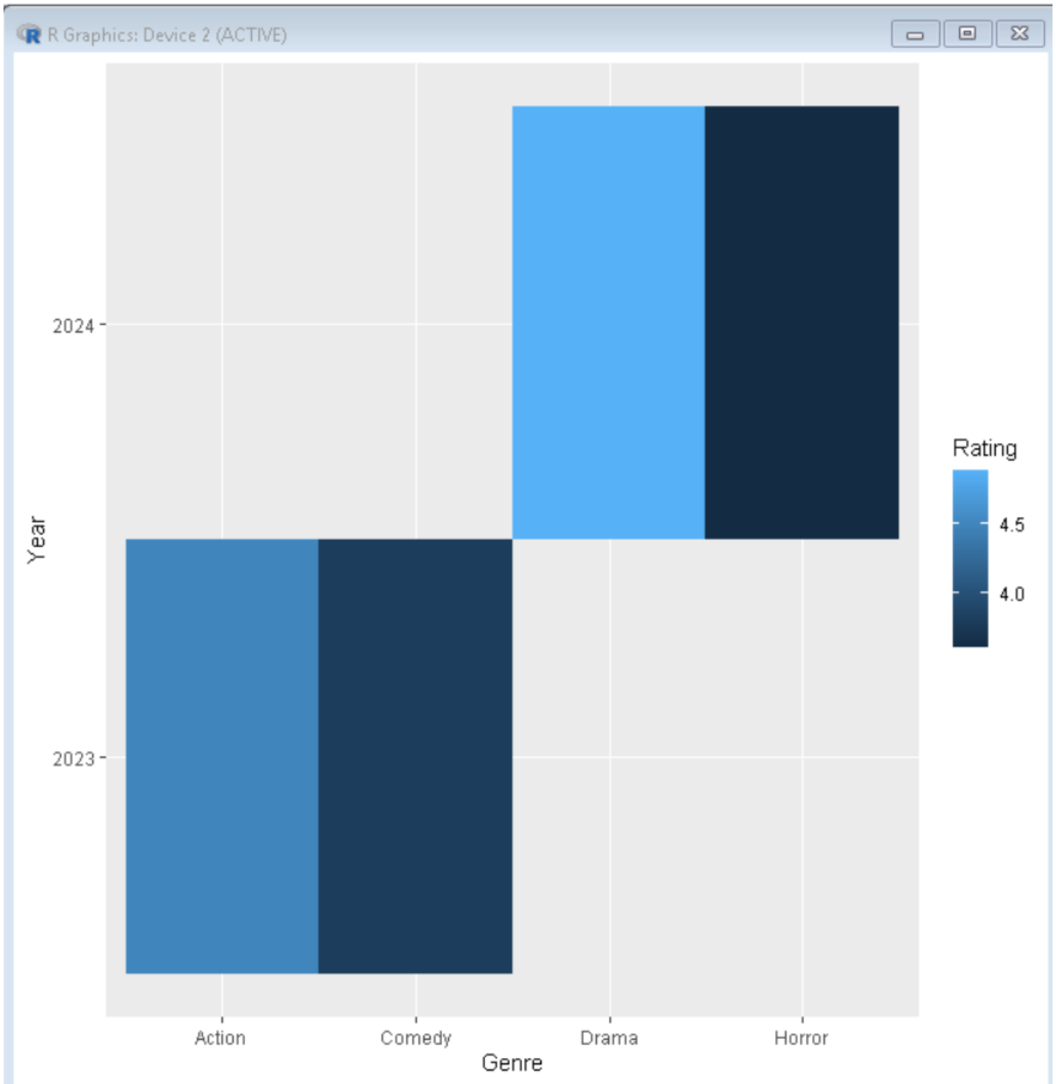
<u>QUESTION 8: SMARTPHONE MARKET COMPARISON</u>

CODE:

```
movies <- data.frame(
  Genre = c("Action","Comedy","Drama","Horror"),
  Year = c("2023","2023","2024","2024"),
  Rating = c(4.5,3.8,4.9,3.6)
)

ggplot(movies,
       aes(x=Genre,
           y=Year,
           fill=Rating)) +
  geom_tile()
```

OUTPUT:



QUESTION 9: COVID-19 STATE-WISE ANALYSIS

CODE:

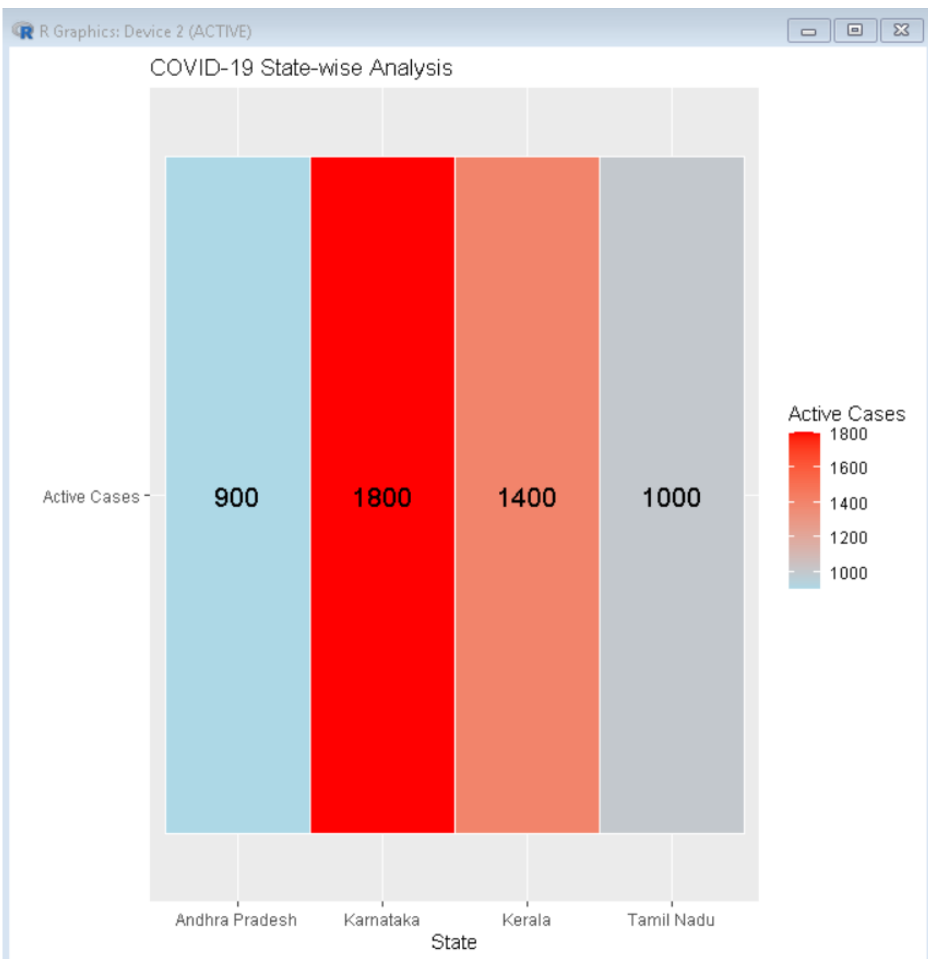
```
library(ggplot2)

covid <- data.frame(
  State = c("Tamil Nadu","Karnataka","Kerala","Andhra Pradesh"),
  Type = "Active Cases",
```

```
  Cases = c(1000,1800,1400,900)
)

ggplot(covid, aes(x = State, y = Type, fill = Cases)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Cases), color = "black", size = 5) +
  scale_fill_gradient(low = "lightblue", high = "red") +
  labs(
    title = "COVID-19 State-wise Analysis",
    x = "State",
    y = "",
    fill = "Active Cases"
  )
```

OUTPUT:



QUESTION 10: IRIS FLOWER CLASSIFICATION

Code:

```
data(iris)

ggplot(iris,
       aes(x=Sepal.Length,
           y=Petal.Length,
           color=Species,
           shape=Species,
           size=Sepal.Width)) +
  geom_point()
```

OUTPUT: